Forcing Reading Group — Exercise Sheet 1

A Quick and Dirty Recap of the Completeness Theorem

"Definition" 1. A *language* L is a set of constant symbols, relation symbols and function symbols, each relation and function symbol with a specified arity. An *L-term* is a well-formed string of L-function symbols, constants and variables. An *atomic L-formula* is either $t_1 = t_2$ for L-terms t_1 and t_2 or $R(t_1, \dots, t_k)$, where $R \in L$ is a k-ary relation symbol. An *L-formula* is a well-formed string consisting of atomic formulas, the logical connectives $\wedge, \vee, \neg$ and the quantifiers $(\forall x), (\exists x)$. An L-sentence is an *L-formula* where every variable is quantified. An *L-theory* is a set of L-sentences.

Definition 2. Let L be a language. An L-structure is a non-empty set S along with an *interpretation* for each symbol in L: An element $c^S \in S$ for every constant symbol $c \in L$, a relation $R^S \subseteq S^k$ for every k-ary relation symbol $R \in L$ and a function $f^S : S^k \to S$ for every function symbol $f \in L$.

"Definition" 3. Let L be a language and S be an L-structure. By recursion on the complexity of an L-formula $\varphi(x_1, \dots, x_k)$, we define $S \models \varphi(s_1, \dots, s_k)$, for $s_1, \dots, s_k \in S$, meaning that φ is true about the elements $s_1, \dots, s_k$ in S.

Given an L-theory T, we say an L-structure M is a *model* of T if for every $\varphi \in T$: $M \models \varphi$. This is denoted $M \models T$. Given another L-theory S, $S \models T$ if whenever $M \models S$, then $M \models T$.

Exercise 4. Find an $\{1, ^{-1}, \cdot\}$-theory T_{group} such that the class of its models are exactly groups.

Remark 5. To give a formal definition of a first order proof is an involved task, we will instead describe which properties a proof should satisfy.

Let L be a language. A proof of an L-sentence φ from a finite set of L-sentences $\varphi_1, \dots, \varphi_k$ is a string. We say φ is provable from an L-theory T, denoted $T \vdash \varphi$, if there is a proof of φ from some finite subset of T. Proofs are sound: If $\{\varphi_1, \dots, \varphi_k\} \vdash \varphi$, then $\varphi_1, \dots, \varphi_k \models \varphi$. For any countable sublanguage of L, it is decidable which strings are valid proofs of a formula from some assumptions.

Proofs are also defined to allow for the proof of the completeness theorem.

Theorem 6 (Completeness)**.** Let L be a language and T be an L-theory. Then,

$$T \models \varphi \Leftrightarrow T \vdash \varphi.$$

Moreover, $T \nvdash \bot$ if and only if $\exists M \models T$ and if L is countable then M can always be found countable.

Exercise 7. Use the completeness theorem to show that $T_{group} \nvdash a \cdot b = b \cdot a$

Exercise 8 (Failure of the finite model property)**.** Find a theory T such that all of its finite models satisfy a sentence φ but $T \nvdash \varphi$.

Nonstandard models

Definition 9. Let $L_{PA} = \{0, 1, +, \cdot, \leq\}$ and let $\mathbb{N}$ be the canonical L_{PA}-structure over the set of natural numbers. We define $\mathrm{Th}(\mathbb{N}) = \{\varphi; \mathbb{N} \models \varphi\}$, this theory is also known as *true arithmetic*.

Theorem 10 (Compactness)**.** Let T be an L-theory, where L is countable. If every finite subset of T has a model, then T has a countable model.

Exercise 11. Show that there is a countable $M \models \mathrm{Th}(\mathbb{N})$ which is not isomorphic to $\mathbb{N}$.

Definition 12. Any $M \models \mathrm{Th}(\mathbb{N})$ which is not isomorphic to $\mathbb{N}$ is called *a nonstandard model* (of true arithmetic).

Exercise 13. Show that $\mathbb{N}$ uniquely embeds into every nonstandard model.

Remark 14. We will always identify $\mathbb{N}$ with its embedding in any nonstandard model, so we will assume that $\mathbb{N}$ is a substructure of every nonstandard model.

Definition 15. Let M be a nonstandard model, the elements $M \setminus \mathbb{N}$ are called *nonstandard* numbers.

Theorem 16 (Cantor)**.** Every countable dense linear order without endpoints is isomorphic to $\mathbb{Q}$.

Exercise 17. Show that the order type of a countable nonstandard model is $\mathbb{N} + \mathbb{Z} \cdot \mathbb{Q}$.

Definition 18. Let M be a nonstandard model, we say that $I \subseteq M$ is a *cut* in M, if $y \leq x \in I \Rightarrow y \in I$ and I is closed under $+1$. We say a cut is *proper* is $I \neq M$.

Exercise 19. Let M be a nonstandard model. Show that no proper cut is definable. Note which axioms of $\mathrm{Th}(\mathbb{N})$ do you use.

Theorem 20 (Overspill I)**.** Let M be a nonstandard model and $\varphi(x)$ an L_{PA}-formula. If for every $s \in \mathbb{N}$: $M \models \varphi(s)$, then there is $n \in M \setminus \mathbb{N}$ such that $M \models (\forall x \leq n)(\varphi(x))$.

Theorem 21 (Overspill II)**.** Let M be a nonstandard model and $\varphi(x)$ an L_{PA}-formula. If for every $s \in \mathbb{N}$, there is $s' \in \mathbb{N}$: $M \models s' > s \wedge \varphi(s')$, then there is $n \in M \setminus \mathbb{N}$ such that $M \models \varphi(n)$. Note which axioms of $\mathrm{Th}(\mathbb{N})$ do you use.